
Session 21 (AIML) - Assignment Questions

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Branch: Artificial Intelligence and Machine Learning (AIML)

Domain: AIML (Artificial Intelligence and Machine Learning)

Github Repo Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments

Github File Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments/tree/main/Session21

Dataset Used: – Kaggle - Ford Car Dataset

Mini Project - Complete Streamlit App

Ford Car Price Prediction Web App

Program Code :

```
1  '''
2  Ford Car Price Prediction Web App
3  '''
4
5  # Import Streamlit for building the web application
6  import streamlit as st
7
8  # Import Pandas for data manipulation
9  import pandas as pd
10
11 # Import Joblib for loading the trained model
12 import joblib
13
14 # -----
15 # Load Saved Model and Preprocessing Objects
16 # -----
17
18 model = joblib.load("LR_ford_car.pkl")
19 scaler = joblib.load("scaler.pkl")
20 encoded_columns = joblib.load("columns.pkl")
21
22 # -----
23 # Configure Streamlit Page
24 # -----
25
26 st.set_page_config(
27     page_title="Ford Car Price Predictor",
28     layout="centered"
29 )
30
31 # -----
32 # Title
33 # -----
34
35 st.title("🚗 Ford Car Price Predictor")
36
37 st.write("Enter the car details below to predict its selling price.")
38
39 st.divider()
40
```

```

41  # -----
42  # Numerical Inputs
43  # -----
44
45  year = st.number_input(
46      "Manufacturing Year",
47      min_value=1990,
48      max_value=2025,
49      value=2018
50  )
51
52  mileage = st.number_input(
53      "Mileage",
54      min_value=0,
55      max_value=300000,
56      value=30000
57  )
58
59  tax = st.number_input(
60      "Road Tax (£)",
61      min_value=0,
62      max_value=600,
63      value=150
64  )
65
66  mpg = st.number_input(
67      "Miles Per Gallon (MPG)",
68      min_value=0.0,
69      max_value=100.0,
70      value=55.0
71  )
72
73  engineSize = st.number_input(
74      "Engine Size (L)",
75      min_value=0.0,
76      max_value=6.0,
77      value=1.5
78  )

```

```

79
80 # -----
81 # Categorical Inputs
82 # -----
83
84 transmission = st.selectbox(
85     "Transmission",
86     [
87         "Automatic",
88         "Manual",
89         "Semi-Auto"
90     ]
91 )
92
93 fuelType = st.selectbox(
94     "Fuel Type",
95     [
96         "Petrol",
97         "Diesel",
98         "Hybrid",
99         "Electric",
100         "Other"
101     ]
102 )
103
104 # -----
105 # Text Input
106 # -----
107
108 model_name = st.text_input(
109     "Car Model"
110 )
111
112 # -----
113 # Predict Button
114 # -----
115
116 predict_button = st.button("Predict Price")
117

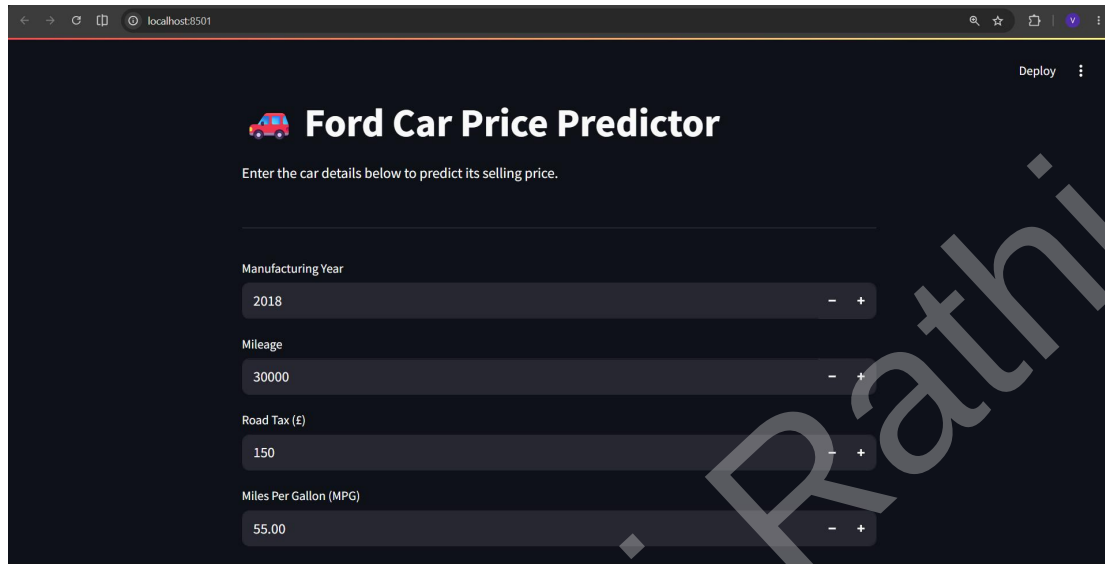
```

```

118 # -----
119 # Prediction
120 # -----
121
122 if predict_button:
123
124     try:
125
126         # Create input DataFrame
127         input_data = pd.DataFrame({
128             "model": [model_name],
129             "year": [year],
130             "transmission": [transmission],
131             "mileage": [mileage],
132             "fuelType": [fuelType],
133             "tax": [tax],
134             "mpg": [mpg],
135             "engineSize": [engineSize]
136         })
137
138         # One-Hot Encoding
139         input_data = pd.get_dummies(input_data)
140
141         # Match columns with training data
142         input_data = input_data.reindex(
143             columns=encoded_columns,
144             fill_value=0
145         )
146
147         # Numerical columns to scale
148         numerical_columns = [
149             "year",
150             "mileage",
151             "tax",
152             "mpg",
153             "engineSize"
154         ]
155
156         # Apply Standard Scaling
157         input_data[numerical_columns] = scaler.transform(
158             input_data[numerical_columns]
159         )
160
161         # Predict Price
162         prediction = model.predict(input_data)
163
164         # Display Prediction
165         st.success(
166             f"🚗 Predicted Car Price: £{prediction[0]:,.2f}"
167         )
168
169     except Exception as e:
170
171         st.error("An error occurred while making the prediction.")
172
173         st.exception(e)

```

Output Screenshot :



The screenshot shows a web browser at localhost:8501 displaying the 'Ford Car Price Predictor' application. The page has a dark theme. At the top right, there is a 'Deploy' button and a menu icon. The main heading is 'Ford Car Price Predictor' with a small car icon. Below the heading, it says 'Enter the car details below to predict its selling price.' There are four input fields with labels and numerical values, each with minus and plus buttons for adjustment:

- Manufacturing Year: 2018
- Mileage: 30000
- Road Tax (£): 150
- Miles Per Gallon (MPG): 55.00



The screenshot shows the same web application with additional input fields. The 'Predict Price' button is highlighted with a red box. Below the button, a green box displays the predicted car price.

Engine Size (L): 1.50

Transmission: Automatic

Fuel Type: Petrol

Car Model: Fiesta

Predict Price

 Predicted Car Price: £12,741.05